

Breaking the Loss Barrier: Hollow-Core Fibers from Physical Limits to Mass Production

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Abstract *This tutorial reviews anti-resonant hollow-core fibers, from structure evolutions and loss reduction to manufacturable long-draw production, and discuss application-oriented trade-offs from intra-data-center links to long-haul networks. ©2026 The Author(s)*

Introduction

As the connectivity radius of AI systems expands, optical interconnects are moving from a supporting role to a limiting system resource. At rack and multi-rack scale, GPU fabrics demand tightly synchronized, high-bandwidth links. Inside large data centers, scale-out traffic already relies on dense optical connectivity. Between data centers, long-reach coherent transmission is the realistic way to sustain the growth of distributed training and inference. Across all of these scenarios, the network is being asked to provide three things at once: low latency, high capacity, and high energy efficiency.

Conventional silica fiber has been astonishingly successful, but its main physical constraints are now plain to see. The propagation speed is set by the refractive index of silica, so optical signals travel at only about $c/1.45$. Channel capacity is ultimately constrained by nonlinear interaction in the glass. Transmission loss is bounded by the Rayleigh-scattering floor, which has driven the best solid-core fibers to around 0.14 dB/km. These limits do not make silica obsolete, but they do mean that future optical networks will benefit from a fundamentally different transmission medium when latency and capacity become co-optimized design variables.

Hollow-core fibers provide exactly that opportunity. By engineering a microstructured silica cladding that confines light to an air core, HCFs inherit several desirable features of free-space propagation while preserving the flexible, long-distance form factor of optical fiber. The result is a transmission channel with near-vacuum propagation speed, very low chromatic dispersion, extremely weak Kerr nonlinearity, and increasingly competitive attenuation. The key question is no longer whether HCFs are interesting, but whether they can scale to useful manufacturing volumes.

Loss Mechanisms

The guiding mechanism is what separates HCFs

most clearly from conventional fibers. In solid-core silica fibers, total internal reflection eliminates confinement loss, leaving Rayleigh scattering in glass as the fundamental attenuation floor. In anti-resonant hollow-core fibers, the core modes are technically leaky modes, since no perfect bound state exists; a small portion of the optical field penetrates into the surrounding membranes. However, the antiresonant condition strongly suppresses this leakage, allowing the mode to propagate over tens to hundreds of kilometers with ultralow loss. For modern anti-resonant fibers, total attenuation can be viewed as the combination of confinement loss, surface-scattering loss, macrobending loss, microbending-induced excess loss [1], and practical perturbation terms associated with fabrication and cabling. Historically, confinement loss dominated and set the early performance ceiling. As cladding designs improved, however, the problem became more subtle: once confinement loss is pushed down sufficiently, surface scattering and bending sensitivity become equally important, and different application scenarios demand different trade-offs among them.

Interstitial-Tube-assisted DNANF (IT-DNANF) for Scalability

HCFs will only matter at network scale if the same structures that produce outstanding optical performance can also be drawn repeatedly, over long lengths, with acceptable yield. This is where IT-DNANF design become crucial. The basic motivation is simple: many elegant low-loss hollow-core designs remain difficult to scale because the draw process is extremely sensitive to pressure balance, surface tension, and fiber tension. Mid-draw contact between elements, gap collapse, and geometric drift can rapidly destroy the intended cladding structure [2].

IT-DNANF addresses this difficulty by adding interstitial tubes that enlarge the structural process window. In practice, this means that the cladding geometry becomes more tolerant of the

competing forces present during drawing. The design preserves the anti-resonant nested-tube concept needed for low confinement loss, while improving gap control, supporting robust single-mode behavior, and reducing the risk of unstable contact events. This is not a cosmetic refinement; it is a shift from “delicate optimization” toward “manufacturable control”.

The reported outcomes are already remarkable. Separate fiber segments have achieved an 83 km single-draw IT-DNANF [3] and low attenuation of 0.038 dB/km [4]. Equally important, pilot-scale production results in monthly outputs of around 1000 km on a single tower have been demonstrated, with weighted-average loss around 0.085 dB/km over large production volumes [4]. Those numbers are not yet the destination, but they establish a credible manufacturing base.

Application-oriented trade-offs and deployments

Different network layers naturally map onto different fiber trade-offs. For intra-rack and intra-data-center links, the main priorities are low bend loss, high density, and ecosystem compatibility, often in the O-band. For data-center interconnects (DCI), low C-band attenuation and good bend tolerance become central. For long-haul and backbone networks, the performance target shifts again: the most critical metrics are ultralow attenuation, low IMI, and sufficient bandwidth, while bend sensitivity can be traded more carefully against absolute loss performance.

This application perspective helps explain why HCF should not be treated as a monolithic technology. A single architecture will not optimize every use case equally well. Instead, the practical roadmap is a portfolio strategy: tight-bend, connector-friendly fibers for short reach; low-loss, low-IMI fibers for DCI; and ultralow-loss, broadband fibers for long-haul systems. In every case, however, the same industrial bottlenecks must be solved—cabling, splicing, HCF–SMF adaptors, reliable characterization, and predictable field installation behavior.

The encouraging news is that deployment has already begun. A China Mobile field trial in a Wuxi data-center environment demonstrated optical-fiber cables with attenuation below 0.14 dB/km [5] and a lot more projects are on going. These examples remain early, but they prove that HCF is no longer confined to the laboratory bench.

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